Type MS · Sound sampling: analogue to digital

Answer 1 of 8 · Question 1 part c.i · 3 marks

2021 NOV · Paper 01 · extract 1 of 4 · find part c.i in this mark scheme extract

1(ai)	0100 1101 100 1101 Award **one** mark for each correct nibble	Must be in the correct order	2
1(a)(ii)	1000 0010 Award **one** mark for each correct nibble	Must be in the correct order	2
1(a)(iii)	Award **one** mark for each of: • C (1) • 6 (1)	Must be in the correct order	2

Type MS · Sound sampling: analogue to digital

Answer 1 of 8 · Question 1 part c.i · 3 marks

2021 NOV · Paper 01 · extract 2 of 4 · find part c.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(b)(i)	400 pixels x 200 pixels x 12 bits / 8 bits per byte Allow expression in word form $\frac{400 \times 200 \times 12}{8}$ 400 x 200 **OR** 80000(1) 12/8 **OR** 1.5 (1)	Accept any other use of 400, 200, 12, 8 that gives the correct answer Allow one mark for 120,000 with no expression	2
1(b)(ii)	Spaces are to help legibility. Award **three** marks for: Line 1 3W 1B 4W Line 2 1W 6B 1W Line 3 1B 2W 1B 1W 1B 2W Line 4 3W 2B 3W Award **two** marks for any three lines correctly encoded Award **one** mark for any two lines correctly encoded	Allow **one** error in the sequence Accept letter before number but must be consistently used	3
1(b)(iii)	Award **one** mark from: • None of the original (image) detail/quality is lost (when the image is stored) (1) • None of the data/information is (permanently) removed (1) • It can be decompressed without losing detail (1)		1

Type MS · Sound sampling: analogue to digital

Answer 1 of 8 · Question 1 part c.i · 3 marks

2021 NOV · Paper 01 · extract 3 of 4 · find part c.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(c)(i)	Award up to **three** marks for a linked description: • set the sample rate/parameters/bit-depth (1) • sample (the analogue sound) (1) • measure the sound amplitude/volume/frequency (1) • give a (binary) value/number for each measurement (1) • store data as sample rate and values / digital signals (1)		3
1(c)(ii)	Award **one** mark from: • sound may be missing some frequencies (1) • some audio information/data is lost/deleted during the (compression) process (1)		1
		Total	**16**

Type MS · Sound sampling: analogue to digital

Answer 1 of 8 · Question 1 part c.i · 3 marks

2021 NOV · Paper 01 · extract 4 of 4 · find part c.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Sound sampling: analogue to digital

Answer 2 of 8 · Question 4 part a.i · 1 mark

2020 NOV · Paper 01 · extract 1 of 4 · find part a.i in this mark scheme extract

4(a)(i)	Digital (signal) / converted analogue (signal) (1)		1

Question Number	Answer	Additional Guidance	Mark
4(a)(ii)	Analogue (signal) / analogue (sound wave) (1)		1

Question Number	Answer	Additional Guidance	Mark
4(a)(iii)	2 Hertz / hertz /Hz (1)	• Do not penalise spelling • Accept 2 samples/cycles per second	1

Question Number	Answer	Additional Guidance	Mark
4(a)(iv)	Any **one** mark for: • The (analogue sound) wave will be represented more accurately / the fidelity/quality of the recording is improved (1) • The file size will increase / more data stored (as each sample takes up disk space) (1)		1

Question Number	Answer	Additional Guidance	Mark
4(a)(v)	• 3 (1) • The range of sampled sound is from 0-6. To store 7 distinct values, you need 3 bits (111 binary = 7 denary) (1) OR graph shows 8 possible values, 3 bits can store 0 - 7 / 8 values (1)		2

Question Number	Answer	Additional Guidance	Mark

12

Type MS · Sound sampling: analogue to digital

Answer 2 of 8 · Question 4 part a.i · 1 mark

2020 NOV · Paper 01 · extract 2 of 4 · find part a.i in this mark scheme extract

4(a)(vi)	Correct binary value 101 (1)	Ignore leading zeros	1

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

Type MS · Sound sampling: analogue to digital

Answer 2 of 8 · Question 4 part a.i · 1 mark

2020 NOV · Paper 01 · extract 3 of 4 · find part a.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
4(b)	819 seconds * 8 bytes * 47,000 hertz 819 x 8 x 47000 819 seconds * 64÷8 bytes * 47,000 hertz 818 x 64 x 47000 8 819 (1) 8 or 64÷8 (1) 47,000 (1) Fully correct expression (1)	Award equivalent expressions	4

Question Number	Answer	Additional Guidance	Mark
4(c)	Any **one** from: • The sound quality may be poorer/lower (for some people) (1) • Cannot get the original back after compression (1)		1

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

Type MS · Sound sampling: analogue to digital

Answer 2 of 8 · Question 4 part a.i · 1 mark

2020 NOV · Paper 01 · extract 4 of 4 · find part a.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Sound sampling: analogue to digital

Answer 3 of 8 · Question 4 part a.ii · 1 mark

2020 NOV · Paper 01 · extract 1 of 4 · find part a.ii in this mark scheme extract

4(a)(i)	Digital (signal) / converted analogue (signal) (1)		1

Question Number	Answer	Additional Guidance	Mark
4(a)(ii)	Analogue (signal) / analogue (sound wave) (1)		1

Question Number	Answer	Additional Guidance	Mark
4(a)(iii)	2 Hertz / hertz /Hz (1)	• Do not penalise spelling • Accept 2 samples/cycles per second	1

Question Number	Answer	Additional Guidance	Mark
4(a)(iv)	Any **one** mark for: • The (analogue sound) wave will be represented more accurately / the fidelity/quality of the recording is improved (1) • The file size will increase / more data stored (as each sample takes up disk space) (1)		1

Question Number	Answer	Additional Guidance	Mark
4(a)(v)	• 3 (1) • The range of sampled sound is from 0-6. To store 7 distinct values, you need 3 bits (111 binary = 7 denary) (1) OR graph shows 8 possible values, 3 bits can store 0 - 7 / 8 values (1)		2

Question Number	Answer	Additional Guidance	Mark

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

Type MS · Sound sampling: analogue to digital

Answer 3 of 8 · Question 4 part a.ii · 1 mark

2020 NOV · Paper 01 · extract 2 of 4 · find part a.ii in this mark scheme extract

4(a)(vi)	Correct binary value 101 (1)	Ignore leading zeros	1

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

Type MS · Sound sampling: analogue to digital

Answer 3 of 8 · Question 4 part a.ii · 1 mark

2020 NOV · Paper 01 · extract 3 of 4 · find part a.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
4(b)	819 seconds * 8 bytes * 47,000 hertz 819 x 8 x 47000 819 seconds * 64÷8 bytes * 47,000 hertz 818 x 64 x 47000 / 8 819 (1) 8 or 64÷8 (1) 47,000 (1) Fully correct expression (1)	Award equivalent expressions	4

Question Number	Answer	Additional Guidance	Mark
4(c)	Any **one** from: • The sound quality may be poorer/lower (for some people) (1) • Cannot get the original back after compression (1)		1

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

Type MS · Sound sampling: analogue to digital

Answer 3 of 8 · Question 4 part a.ii · 1 mark

2020 NOV · Paper 01 · extract 4 of 4 · find part a.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Sound sampling: analogue to digital

Answer 4 of 8 · Question 4 part a.iii · 1 mark

2020 NOV · Paper 01 · extract 1 of 4 · find part a.iii in this mark scheme extract

4(a)(i)	Digital (signal) / converted analogue (signal) (1)		1

Question Number	Answer	Additional Guidance	Mark
4(a)(ii)	Analogue (signal) / analogue (sound wave) (1)		1

Question Number	Answer	Additional Guidance	Mark
4(a)(iii)	2 Hertz / hertz /Hz (1)	• Do not penalise spelling • Accept 2 samples/cycles per second	1

Question Number	Answer	Additional Guidance	Mark
4(a)(iv)	Any **one** mark for: • The (analogue sound) wave will be represented more accurately / the fidelity/quality of the recording is improved (1) • The file size will increase / more data stored (as each sample takes up disk space) (1)		1

Question Number	Answer	Additional Guidance	Mark
4(a)(v)	• 3 (1) • The range of sampled sound is from 0-6. To store 7 distinct values, you need 3 bits (111 binary = 7 denary) (1) OR graph shows 8 possible values, 3 bits can store 0 - 7 / 8 values (1)		2

Question Number	Answer	Additional Guidance	Mark

Type MS · Sound sampling: analogue to digital

Answer 4 of 8 · Question 4 part a.iii · 1 mark

2020 NOV · Paper 01 · extract 2 of 4 · find part a.iii in this mark scheme extract

4(a)(vi)	Correct binary value 101 (1)	Ignore leading zeros	1

Type MS · Sound sampling: analogue to digital

Answer 4 of 8 · Question 4 part a.iii · 1 mark

2020 NOV · Paper 01 · extract 3 of 4 · find part a.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
4(b)	819 seconds * 8 bytes * 47,000 hertz 819 x 8 x 47000 819 seconds * 64÷8 bytes * 47,000 hertz 818 x 64 x 47000 8 819 (1) 8 or 64÷8 (1) 47,000 (1) Fully correct expression (1)	Award equivalent expressions	4

Question Number	Answer	Additional Guidance	Mark
4(c)	Any **one** from: • The sound quality may be poorer/lower (for some people) (1) • Cannot get the original back after compression (1)		1

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Type MS · Sound sampling: analogue to digital

Answer 4 of 8 · Question 4 part a.iii · 1 mark

2020 NOV · Paper 01 · extract 4 of 4 · find part a.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Sound sampling: analogue to digital

Answer 5 of 8 · Question 4 part a.iv · 1 mark

2020 NOV · Paper 01 · extract 1 of 4 · find part a.iv in this mark scheme extract

4(a)(i)	Digital (signal) / converted analogue (signal) (1)		1

Question Number	Answer	Additional Guidance	Mark
4(a)(ii)	Analogue (signal) / analogue (sound wave) (1)		1

Question Number	Answer	Additional Guidance	Mark
4(a)(iii)	2 Hertz / hertz /Hz (1)	• Do not penalise spelling • Accept 2 samples/cycles per second	1

Question Number	Answer	Additional Guidance	Mark
4(a)(iv)	Any **one** mark for: • The (analogue sound) wave will be represented more accurately / the fidelity/quality of the recording is improved (1) • The file size will increase / more data stored (as each sample takes up disk space) (1)		1

Question Number	Answer	Additional Guidance	Mark
4(a)(v)	• 3 (1) • The range of sampled sound is from 0-6. To store 7 distinct values, you need 3 bits (111 binary = 7 denary) (1) OR graph shows 8 possible values, 3 bits can store 0 - 7 / 8 values (1)		2

Question Number	Answer	Additional Guidance	Mark

Type MS · Sound sampling: analogue to digital

Answer 5 of 8 · Question 4 part a.iv · 1 mark

2020 NOV · Paper 01 · extract 2 of 4 · find part a.iv in this mark scheme extract

4(a)(vi)	Correct binary value 101 (1)	Ignore leading zeros	1

Type MS · Sound sampling: analogue to digital

Answer 5 of 8 · Question 4 part a.iv · 1 mark

2020 NOV · Paper 01 · extract 3 of 4 · find part a.iv in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
4(b)	819 seconds * 8 bytes * 47,000 hertz 819 x 8 x 47000 819 seconds * 64÷8 bytes * 47,000 hertz 818 x 64 x 47000 8 819 (1) 8 or 64÷8 (1) 47,000 (1) Fully correct expression (1)	Award equivalent expressions	4

Question Number	Answer	Additional Guidance	Mark
4(c)	Any **one** from: • The sound quality may be poorer/lower (for some people) (1) • Cannot get the original back after compression (1)		1

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Type MS · Sound sampling: analogue to digital

Answer 5 of 8 · Question 4 part a.iv · 1 mark

2020 NOV · Paper 01 · extract 4 of 4 · find part a.iv in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Sound sampling: analogue to digital

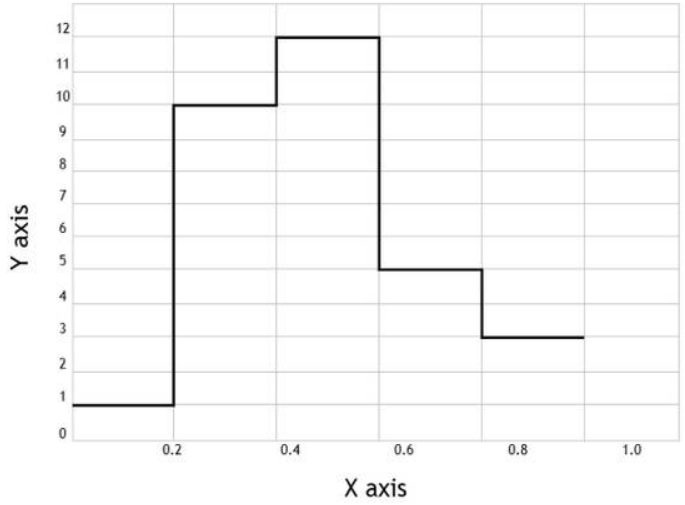
Answer 6 of 8 · Question 3 part a.i · 3 marks

2022 JUN · Paper 01 · extract 1 of 4 · find part a.i in this mark scheme extract

<table>
<tr>
<td>3(a)(i)</td>
<td>Award one mark for each of:

• At least four conversions plotted at the correct amplitude (1)
• Correct start point (1) 0 – 1 digital. 0 – 0 analogue
• Digital sound wave drawn (1) using candidate's plots

<table>
<tr><th>Sample number</th><th>Denary value</th></tr>
<tr><td>1</td><td>1</td></tr>
<tr><td>2</td><td>10</td></tr>
<tr><td>3</td><td>12</td></tr>
<tr><td>4</td><td>5</td></tr>
<tr><td>5</td><td>3</td></tr>
</table>

 
</td>
<td>MP1 does not have to start at 0.2 - 1</td>
<td>3</td>
</tr>
<tr>
<td>3(a)(ii)</td>
<td>Award one mark for:

• Time (1)
• Sample interval/period (1)
• Seconds (1)</td>
<td></td>
<td>1</td>
</tr>
<tr>
<td>3(a)(iii)</td>
<td>Award one mark for:

• Amplitude / sound level / volume (1)
• Metres/centimetres/nanometres (1)
• m/cm/nm (1)</td>
<td></td>
<td>1</td>
</tr>
</table>

Type MS · Sound sampling: analogue to digital

Answer 6 of 8 · Question 3 part a.i · 3 marks

2022 JUN · Paper 01 · extract 2 of 4 · find part a.i in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>3(b)(i)</td><td>Award one mark from:
Transfer time would be longer (1)
Larger file size / less compression (1)
Takes up more storage space (1)
Uses more of her data allocation (1)</td><td></td><td>1</td></tr>
<tr><td>3(b)(ii)</td><td>Award up to two marks for a linked explanation such as:
Storage capacity can be scaled up and down (1) so no need to buy more secondary storage / only pay for what is used/needed (1)
Will have the files all in one place (1) rather than scattered across many secondary storage devices (1)
Files can be uploaded/downloaded anytime/anywhere/on any device (1) so long as there is an internet connection (1)
Can be set up to automatically backup / synchronise with mobile devices (1) therefore if any files/devices are lost/stolen her files will be available on the server (1)</td><td>Do not award a mark for cheaper without expansion.</td><td>2</td></tr>
<tr><td>3(b)(iii)</td><td>Award one mark from:
Storage host could be targeted by hackers (1)
Alyssa has less control over her files (1)
An untrustworthy employee (of the cloud storage provider) could steal her files (1)
Reliant on the storage provider for security / safekeeping (1)
Files could be intercepted/corrupted during upload/download (1)</td><td></td><td>1</td></tr>
<tr><td>3(b)(iv)</td><td>Award one mark for each of:<table><tr><th>URL component</th><th>Description</th></tr><tr><td>https</td><td>Protocol / scheme (1)</td></tr><tr><td>www.cloudisfab.com</td><td>Domain (name) / host / name of website (1)</td></tr><tr><td>re12</td><td>Folder/directory (on the website) / path / part of path(1)</td></tr></table></td><td></td><td>4</td></tr>
</table>

Type MS · Sound sampling: analogue to digital

Answer 6 of 8 · Question 3 part a.i · 3 marks

2022 JUN · Paper 01 · extract 3 of 4 · find part a.i in this mark scheme extract

	ru2.mp3	File/media/resource wanted (on the website)/ path / part of path (1)		
			Total for question 3	**13**

Type MS · Sound sampling: analogue to digital

Answer 6 of 8 · Question 3 part a.i · 3 marks

2022 JUN · Paper 01 · extract 4 of 4 · find part a.i in this mark scheme extract

Question Number	Answer (flow chart replaced)	Additional Guidance	Mark

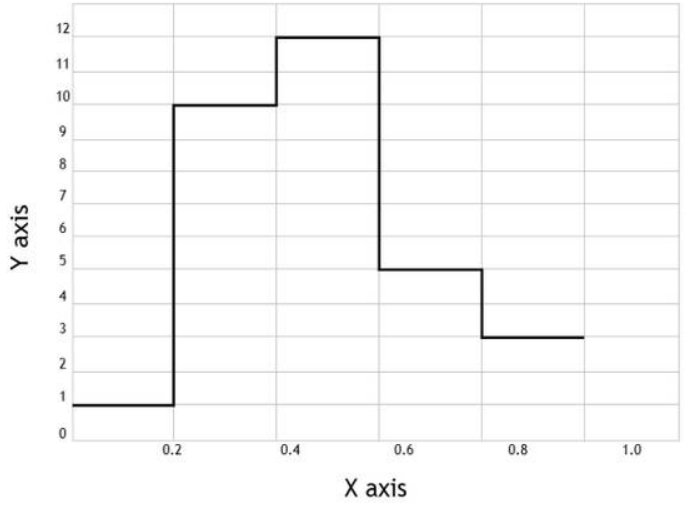
Type MS · Sound sampling: analogue to digital

Answer 7 of 8 · Question 3 part a.ii · 1 mark

2022 JUN · Paper 01 · extract 1 of 4 · find part a.ii in this mark scheme extract

<table>
<tr><td>3(a)(i)</td><td>Award one mark for each of:

• At least four conversions plotted at the correct amplitude (1)
• Correct start point (1) 0 – 1 digital. 0 – 0 analogue
• Digital sound wave drawn (1) using candidate's plots

<table><tr><th>Sample number</th><th>Denary value</th></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>10</td></tr><tr><td>3</td><td>12</td></tr><tr><td>4</td><td>5</td></tr><tr><td>5</td><td>3</td></tr></table>
 
</td><td>MP1 does not have to start at 0.2 - 1</td><td>3</td></tr>
<tr><td>3(a)(ii)</td><td>Award one mark for:

• Time (1)
• Sample interval/period (1)
• Seconds (1)</td><td></td><td>1</td></tr>
<tr><td>3(a)(iii)</td><td>Award one mark for:

• Amplitude / sound level / volume (1)
• Metres/centimetres/nanometres (1)
• m/cm/nm (1)</td><td></td><td>1</td></tr>
</table>

Type MS · Sound sampling: analogue to digital

Answer 7 of 8 · Question 3 part a.ii · 1 mark

2022 JUN · Paper 01 · extract 2 of 4 · find part a.ii in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>3(b)(i)</td><td>Award one mark from:
Transfer time would be longer (1)
Larger file size / less compression (1)
Takes up more storage space (1)
Uses more of her data allocation (1)</td><td></td><td>1</td></tr>
<tr><td>3(b)(ii)</td><td>Award up to two marks for a linked explanation such as:
Storage capacity can be scaled up and down (1) so no need to buy more secondary storage / only pay for what is used/needed (1)
Will have the files all in one place (1) rather than scattered across many secondary storage devices (1)
Files can be uploaded/downloaded anytime/anywhere/on any device (1) so long as there is an internet connection (1)
Can be set up to automatically backup / synchronise with mobile devices (1) therefore if any files/devices are lost/stolen her files will be available on the server (1)</td><td>Do not award a mark for cheaper without expansion.</td><td>2</td></tr>
<tr><td>3(b)(iii)</td><td>Award one mark from:
Storage host could be targeted by hackers (1)
Alyssa has less control over her files (1)
An untrustworthy employee (of the cloud storage provider) could steal her files (1)
Reliant on the storage provider for security / safekeeping (1)
Files could be intercepted/corrupted during upload/download (1)</td><td></td><td>1</td></tr>
<tr><td>3(b)(iv)</td><td>Award one mark for each of:<table><tr><th>URL component</th><th>Description</th></tr><tr><td>https</td><td>Protocol / scheme (1)</td></tr><tr><td>www.cloudisfab.com</td><td>Domain (name) / host / name of website (1)</td></tr><tr><td>re12</td><td>Folder/directory (on the website) / path / part of path(1)</td></tr></table></td><td></td><td>4</td></tr>
</table>

Type MS · Sound sampling: analogue to digital

Answer 7 of 8 · Question 3 part a.ii · 1 mark

2022 JUN · Paper 01 · extract 3 of 4 · find part a.ii in this mark scheme extract

	ru2.mp3	File/media/resource wanted (on the website)/ path / part of path (1)		
			Total for question 3	**13**

Type MS · Sound sampling: analogue to digital

Answer 7 of 8 · Question 3 part a.ii · 1 mark

2022 JUN · Paper 01 · extract 4 of 4 · find part a.ii in this mark scheme extract

Question Number	Answer (flow chart replaced)	Additional Guidance	Mark

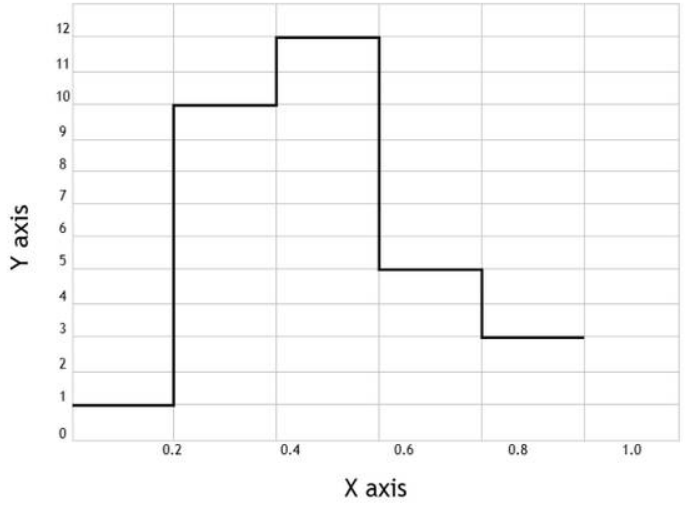
Type MS · Sound sampling: analogue to digital

Answer 8 of 8 · Question 3 part a.iii · 1 mark

2022 JUN · Paper 01 · extract 1 of 4 · find part a.iii in this mark scheme extract

<table>
<tr><td>3(a)(i)</td><td>Award one mark for each of:

• At least four conversions plotted at the correct amplitude (1)
• Correct start point (1) 0 – 1 digital. 0 – 0 analogue
• Digital sound wave drawn (1) using candidate's plots

<table><tr><th>Sample number</th><th>Denary value</th></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>10</td></tr><tr><td>3</td><td>12</td></tr><tr><td>4</td><td>5</td></tr><tr><td>5</td><td>3</td></tr></table>
 
</td><td>MP1 does not have to start at 0.2 - 1</td><td>3</td></tr>
<tr><td>3(a)(ii)</td><td>Award one mark for:

• Time (1)
• Sample interval/period (1)
• Seconds (1)</td><td></td><td>1</td></tr>
<tr><td>3(a)(iii)</td><td>Award one mark for:

• Amplitude / sound level / volume (1)
• Metres/centimetres/nanometres (1)
• m/cm/nm (1)</td><td></td><td>1</td></tr>
</table>

Type MS · Sound sampling: analogue to digital

Answer 8 of 8 · Question 3 part a.iii · 1 mark

2022 JUN · Paper 01 · extract 2 of 4 · find part a.iii in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>3(b)(i)</td><td>Award one mark from:
Transfer time would be longer (1)
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Takes up more storage space (1)
Uses more of her data allocation (1)</td><td></td><td>1</td></tr>
<tr><td>3(b)(ii)</td><td>Award up to two marks for a linked explanation such as:
Storage capacity can be scaled up and down (1) so no need to buy more secondary storage / only pay for what is used/needed (1)
Will have the files all in one place (1) rather than scattered across many secondary storage devices (1)
Files can be uploaded/downloaded anytime/anywhere/on any device (1) so long as there is an internet connection (1)
Can be set up to automatically backup / synchronise with mobile devices (1) therefore if any files/devices are lost/stolen her files will be available on the server (1)</td><td>Do not award a mark for cheaper without expansion.</td><td>2</td></tr>
<tr><td>3(b)(iii)</td><td>Award one mark from:
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Alyssa has less control over her files (1)
An untrustworthy employee (of the cloud storage provider) could steal her files (1)
Reliant on the storage provider for security / safekeeping (1)
Files could be intercepted/corrupted during upload/download (1)</td><td></td><td>1</td></tr>
<tr><td>3(b)(iv)</td><td>Award one mark for each of:<table><tr><th>URL component</th><th>Description</th></tr><tr><td>https</td><td>Protocol / scheme (1)</td></tr><tr><td>www.cloudisfab.com</td><td>Domain (name) / host / name of website (1)</td></tr><tr><td>re12</td><td>Folder/directory (on the website) / path / part of path(1)</td></tr></table></td><td></td><td>4</td></tr>
</table>

Type MS · Sound sampling: analogue to digital

Answer 8 of 8 · Question 3 part a.iii · 1 mark

2022 JUN · Paper 01 · extract 3 of 4 · find part a.iii in this mark scheme extract

	ru2.mp3	File/media/resource wanted (on the website)/ path / part of path (1)		
			Total for question 3	**13**

Type MS · Sound sampling: analogue to digital

Answer 8 of 8 · Question 3 part a.iii · 1 mark

2022 JUN · Paper 01 · extract 4 of 4 · find part a.iii in this mark scheme extract

Question Number	Answer (flow chart replaced)	Additional Guidance	Mark